Although we do not have the technology to probe such minute distance scales or to produce such scorching temperatures, since 1974 experimentalists have significantly refined the measured strengths of the three nongravitational forces under everyday conditions. These data—the starting points for the three force-strength curves in Figure 7.1—are the input data for the quantum-mechanical extrapolations of Georgi, Quinn, and Weinberg. In 199 1, Ugo Amaldi of CERN, Wim de Boer and Hermann Fürstenau of the University of Karlsruhe, Germany, recalculated the Georgi, Quinn, and Weinberg extrapolations making use of these experimental refinements and showed two significant things. First, the strengths of the three nongravitational forces almost agree, but not quite at tiny distance scales (equivalently, high energy/high temperature) as shown in Figure 7.2. Second, this tiny but undeniable discrepancy in their strengths vanishes if supersymmetry is incorporated. The reason is that the new superpartner particles required by supersymmetry contribute additional quantum fluctuations, and these fluctuations are just right to nudge the strengths of the forces to converge with one another.

To many physicists, it is extremely difficult to believe that nature would choose the forces so that they almost, but not quite, have strengths that microscopically unify—microscopically become equal. It’s like putting together a jigsaw puzzle in which the final piece is slightly misshapen and won’t cleanly fit into its appointed position. Supersymmetry deftly refines its shape so that all pieces firmly lock into place.

Another aspect of this latter realization is that it provides a possible answer to the question, Why haven’t we discovered any of the superpartner particles? The calculations that lead to the convergence of the force strengths, as well as other considerations studied by a number of physicists, indicate that the superpartner particles must be a good deal heavier than the known particles. Although no definitive predictions can be made, studies show that the superpartner particles might be a thousand times as massive as a proton, if not heavier. As even our state-of-the-art accelerators cannot quite reach such energies, this provides an explanation for why these particles have not, as yet, been discovered. In Chapter 9, we will return to a discussion of the